

YTL*AI LABS



UM Hackathon 2026

PEKOM



UNIVERSITI
MALAYA

Z.AI

PeopleGraph

**The AI Decision Engine & Digital
Twin for Malaysian SMEs.**

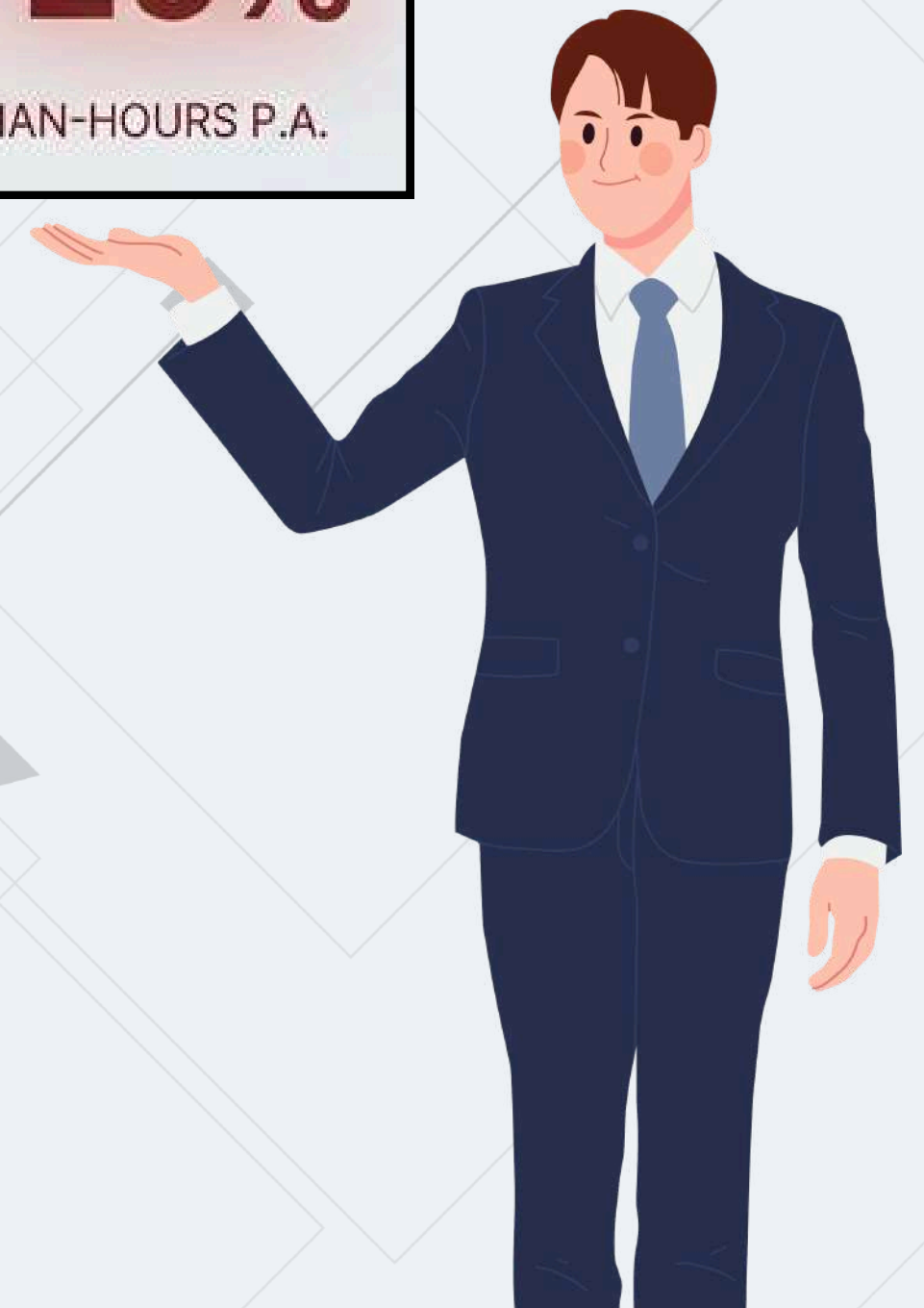
*Bridging the gap between raw workforce data
and revenue-generating intelligence.*

Problem Statement

Value of workers are undefined

Only 10%-15% of net margin for the entire SME

HR Mismanagement



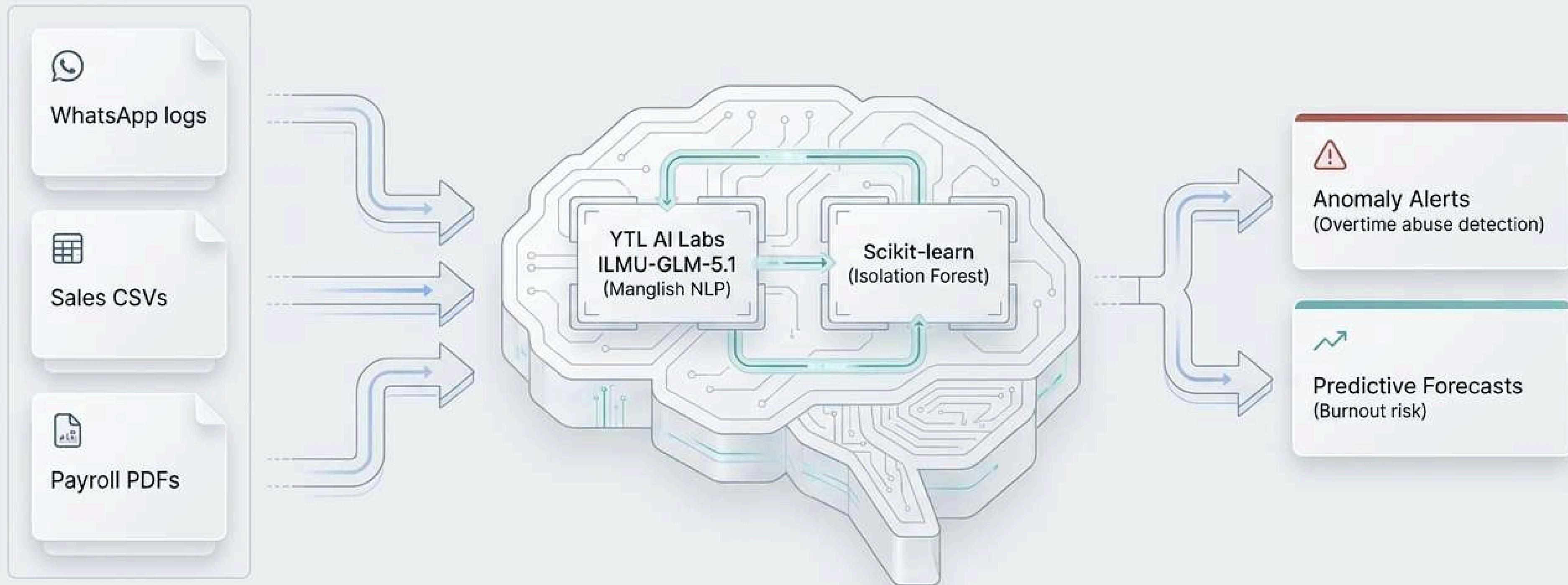
Solution



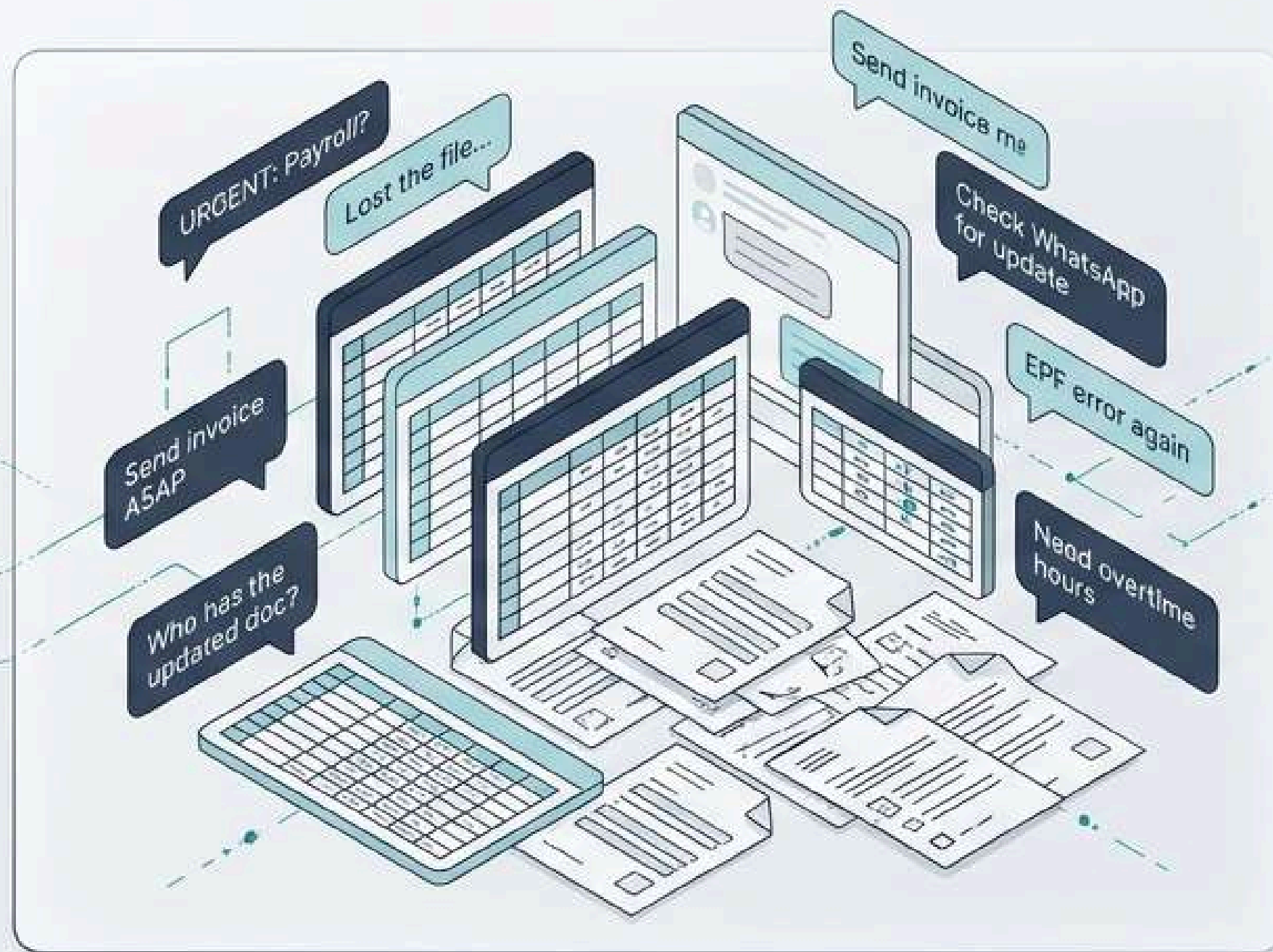
PeopleGraph

The Intelligence Engine: Processing the Unstructured

Purpose-built for the reality of Malaysian business communication (code-switching, informal reporting) to surface hidden operational bottlenecks.



Malaysian SMEs operate in the dark.




Compliance Penalties
(Akta Kerja 1955)

RM

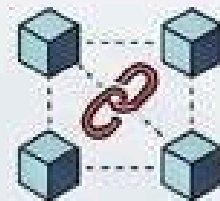
RM [EST. VALUE] P.A.


Hidden Labour Waste

15-20%

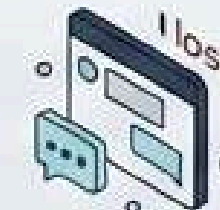
15-20% MAN-HOURS P.A.

Data Silos



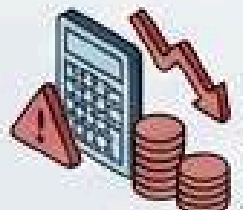
Decisions made on gut-feel, trapping valuable operational data in disconnected spreadsheets.

The Unstructured Trap



Critical workforce intelligence (burnout, bottlenecks) is lost in daily WhatsApp logs.

Compliance Risks



Manual EPF/SOCSSO calculation errors actively drain cash flow through statutory fines.

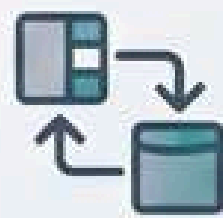
The Digital Twin: From Static Data to Decision Intelligence

PeopleGraph converts fragmented operational records into a live, interactive map that **directly links human activity to cost and revenue outcomes**.



AI Predictive Analytics:

Forecasting hiring impacts before capital is spent.



Unstructured Ingestion:

Making sense of informal Manglish communications.



Automated Compliance:

100% accurate, LHDN-formatted statutory calculations.

The Executive Intelligence Dashboard

Labour Efficiency Ratio (LER):
RM 4.20 earned per RM 1 spent.

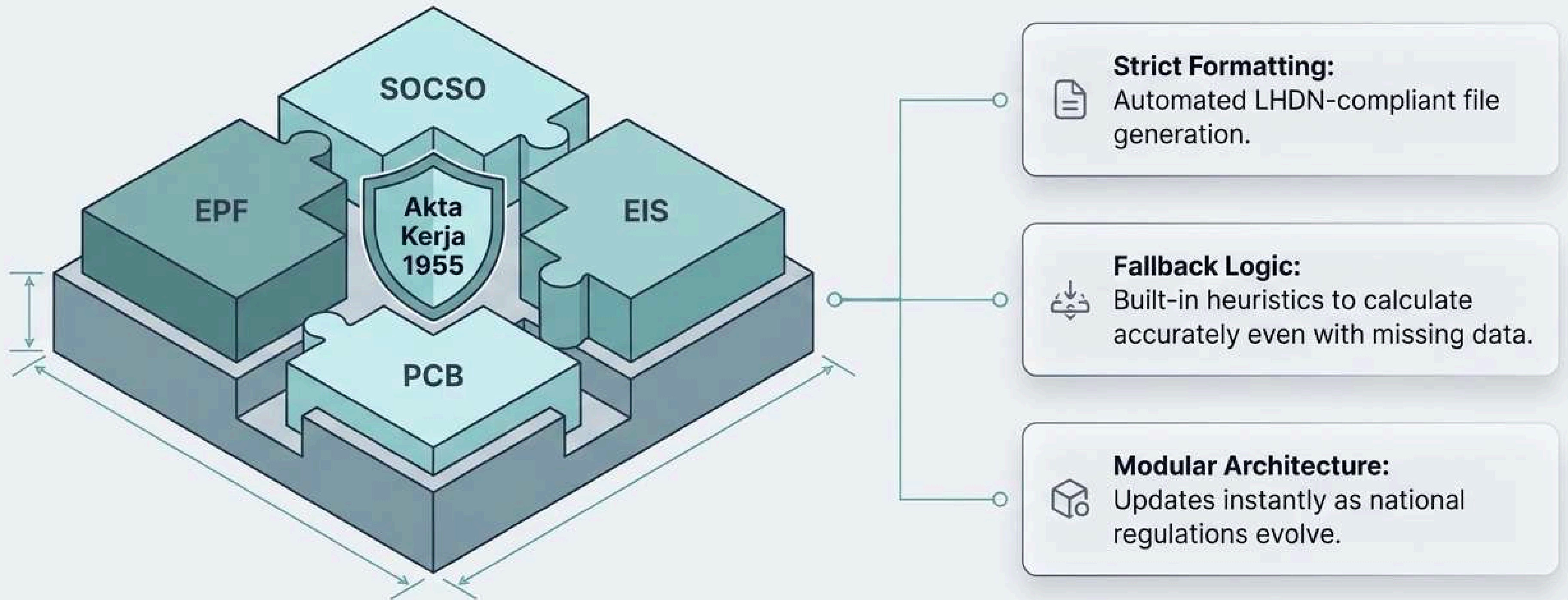


Raya Peak Readiness:
Fulfillment capacity capped at 82%.

The 3-Action AI Framework

- **Automate:** Deploy AI OCR for waybills (Saves RM 4,500/mo).
- **Hire:** Add 1 Supply Chain Manager to unlock 30% capacity.
- **Outsource:** Shift seasonal design work to reduce RM 60k/yr overhead.

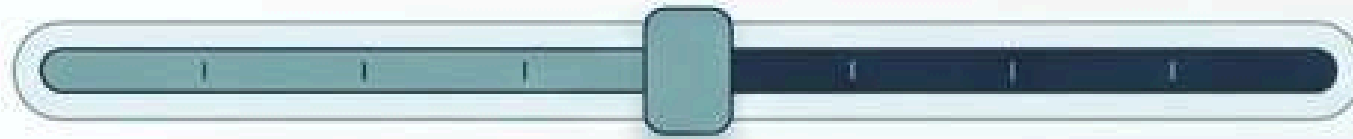
Automated Statutory Compliance



Low-friction compliance automation acts as the daily-use entry hook, opening the door for premium AI upsells.

Predictive Analytics & Shock Simulation

Cuti Peristiwa (Sudden Public Holiday)



Scenario: Modelling the financial impact of a sudden public holiday on warehouse operations.

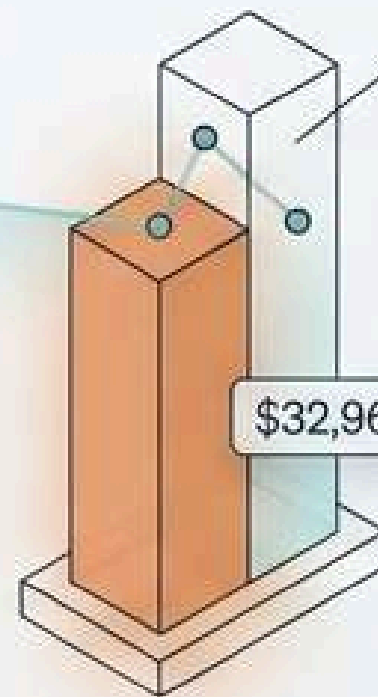
Option A:
Statutory Wage Penalty
(Overtime multipliers)

\$36,134.FM



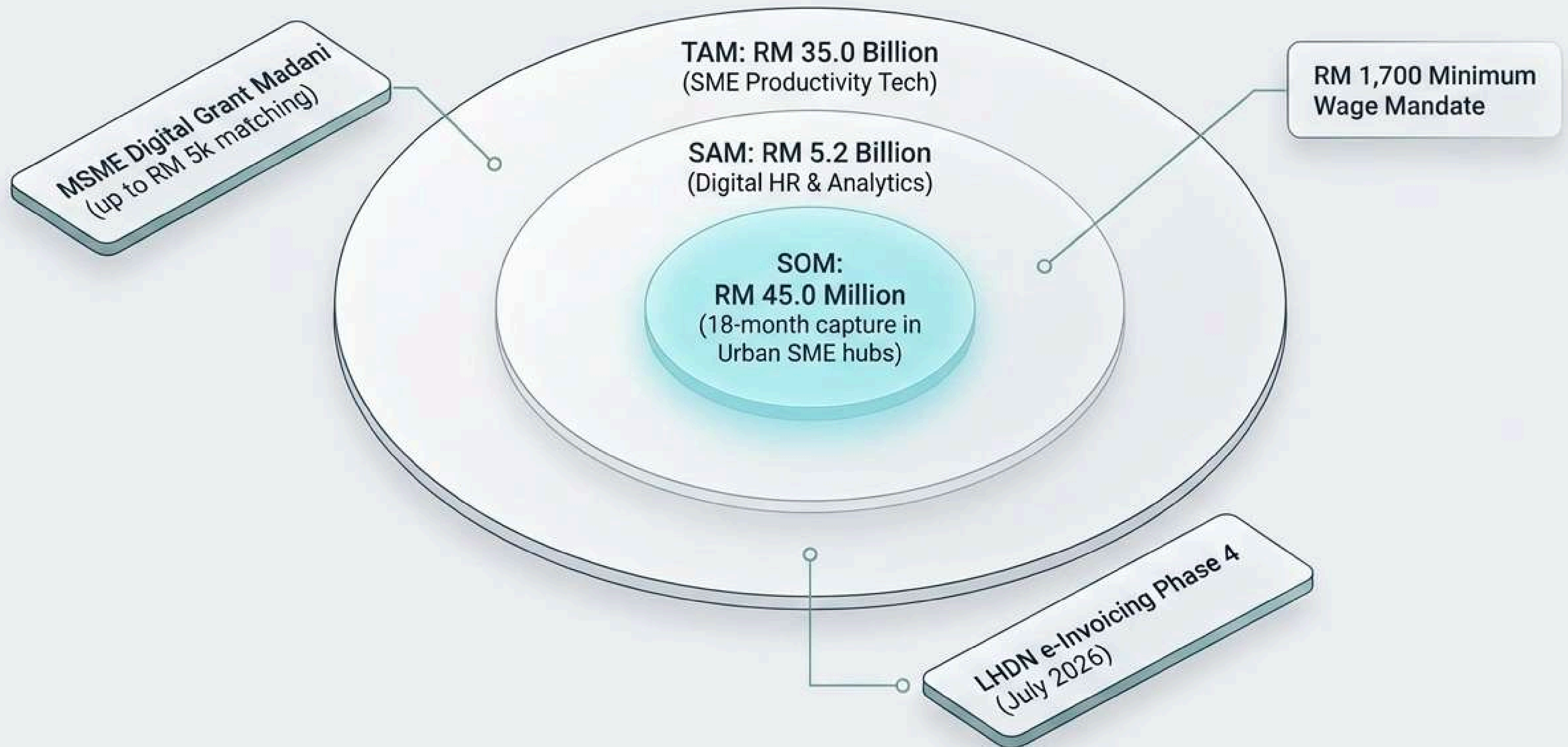
Option B:
Warehouse SLA Impact
(Cost of delayed shipments)

\$32,961.FM

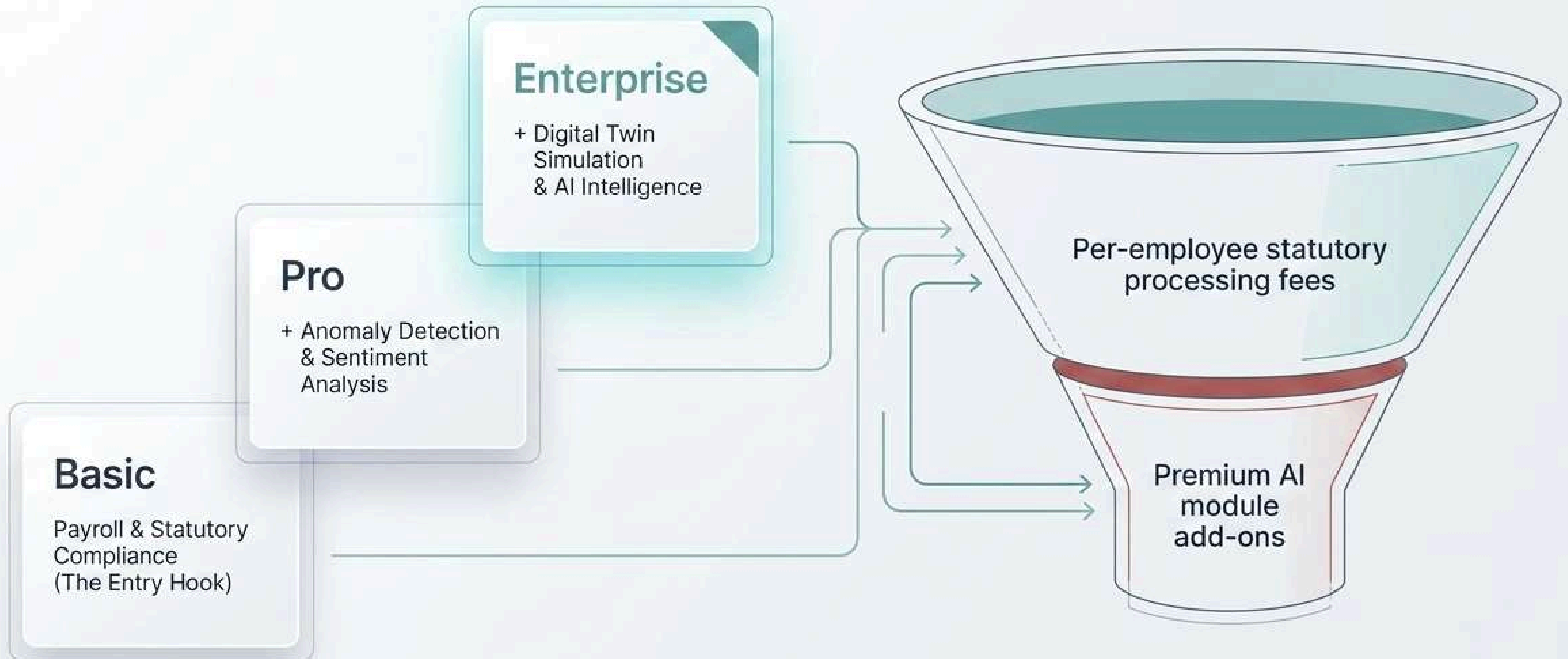


Enables SMEs to run 30-day revenue projections and model operational risks before committing capital.

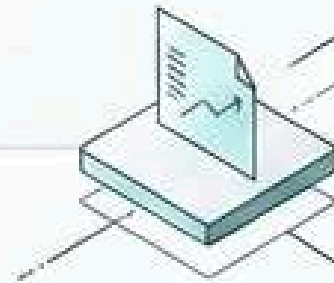
Market Sizing & The Digitalisation Wave



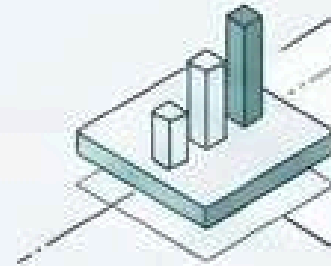
SaaS Monetisation Strategy



3-Year Financial Trajectory



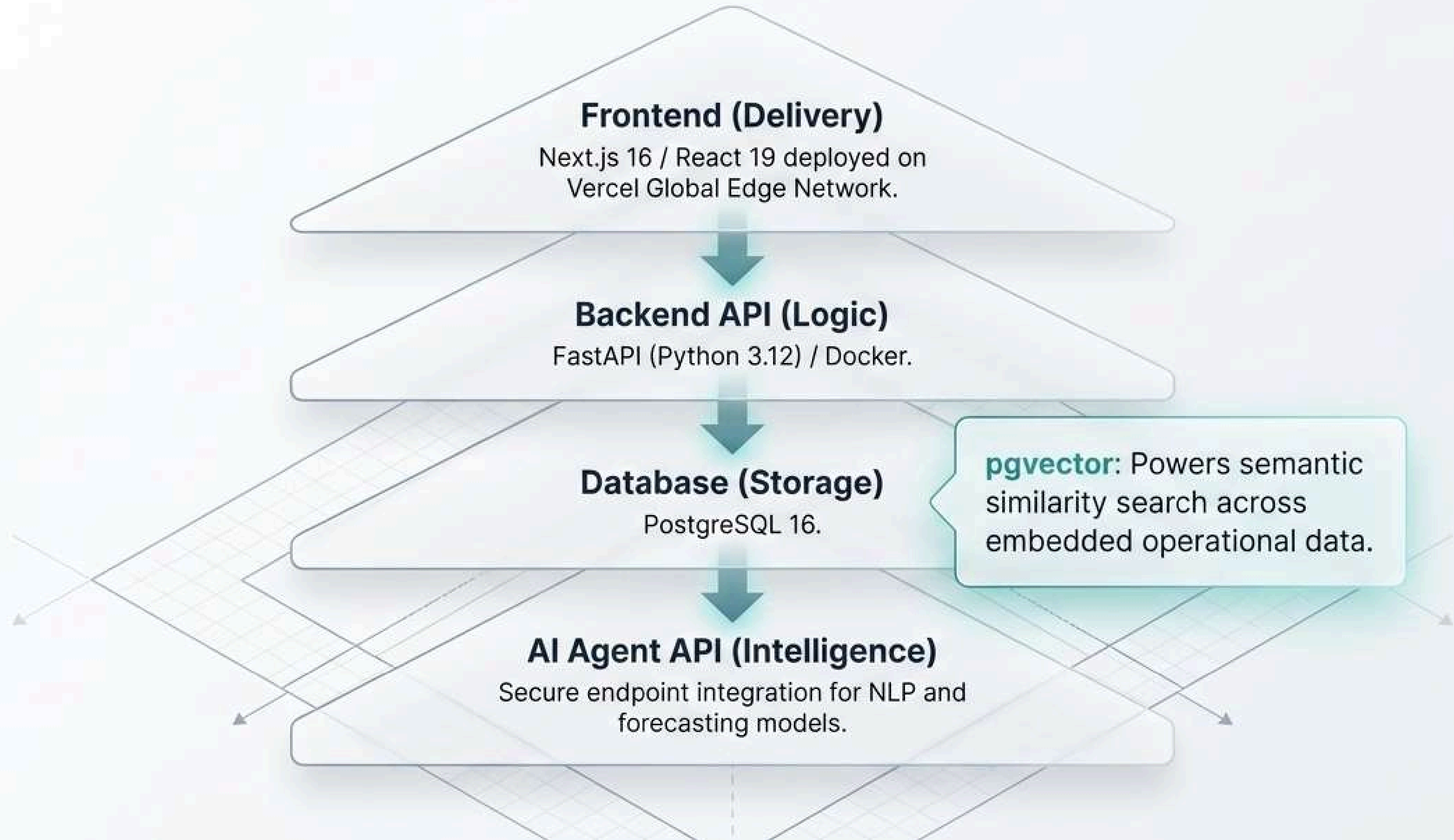
Month 12:
Break-even Achieved



Year 3: Scale across 5
major urban SME hubs



Enterprise-Grade System Architecture





PeopleGraph

An On-Demand Skill-to-Revenue Orchestrator
& AI Decision Engine for Malaysian SMEs

TECH STACK



DATA TO VALUE

Map workforce activities
to revenue outcomes



AI DECISION ENGINE

Predict, simulate &
recommend actions



COMPLIANT & SECURE

Built for Malaysian SMBs
with statutory compliance

CLIENT



NEXT.js

Next.js 16
(React 19)



Dashboards



Value Graph
(D3.js)



Strategic
Insight Panel



Statutory
Compliance View



Analytics &
Simulations



Tailwind CSS 4
TypeScript

HTTPS / CORS

BACKEND



FastAPI

Python 3.10+
Async REST API

API ROUTERS



Employees
/api/v1/employees



Compliance
/api/v1/compliance



Dashboard
/api/v1/dashboard



AI Analytics
/api/v1/ai



SERVICES LAYER
Business Logic
& Domain Services



SCHEMAS
Pydantic Models
(Validation)

DATABASE



PostgreSQL 14+
with pgvector

SQLAlchemy (Async)
Alembic (Migrations)

DATA & AI



UNSTRUCTURED DATA INGESTION



WhatsApp
Logs



Sales
CSV



Payroll
PDF

No Manual KPI Entry

AI & ANALYTICS



scikit-learn
(Ensemble Models)



pandas
(Data Processing)

Forecasting • Simulations • Optimization

NLP & SENTIMENT



YTL AI Labs
ILMU-GLM-5.1



Sentiment Analysis
& Insights

Employee Feedback • Onboarding

SIMULATIONS



Cuti Peristiwa
Shock Simulation



Hiring ROI &
EA1955 Risk

What-if • Risk • Impact

INFRASTRUCTURE & DEVOPS



Docker
Containerization



Docker Compose
Orchestration



GitHub
Version Control



Nginx
Reverse Proxy



CORS
Middleware



Vercel
(Frontend Hosting)

EXTERNAL INTEGRATIONS



EPF
(KWSP)



SOC SO



EIS
(PERKESO)



LHDN
(PCB)



YTL AI Labs
ILMU-GLM-5.1
LLM API



Third-Party
APIs & Services



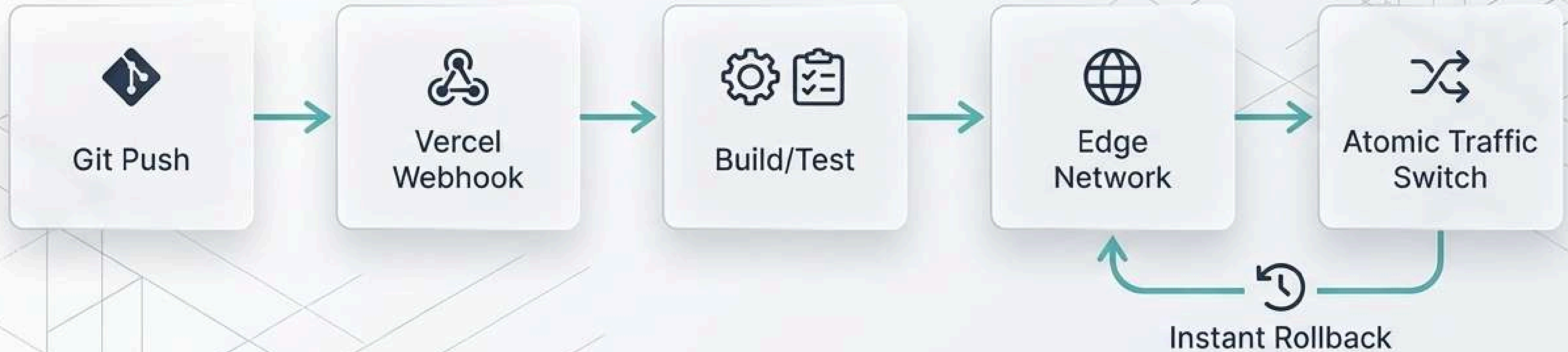
OUTCOMES

- ✓ Map People to Revenue
- ✓ Predict & Optimize Workforce Impact
- ✓ Automate Statutory Compliance
- ✓ Reduce Risk & Operational Shock
- ✓ Drive Strategic Decisions

LEGEND

- Application Layer
- Data & AI Layer
- External / Ecosystem

Zero-Downtime CI/CD Pipeline



Speed



2-4 minute total pipeline duration.

Stability



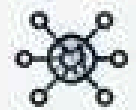
Blue-Green deployment strategy.

Safety



Instant 30-day automated rollback capabilities.

Security & Data Integrity



Access & Edge

HTTPS enforced globally,
JWT Authentication, Strict
CORS allow-listing.



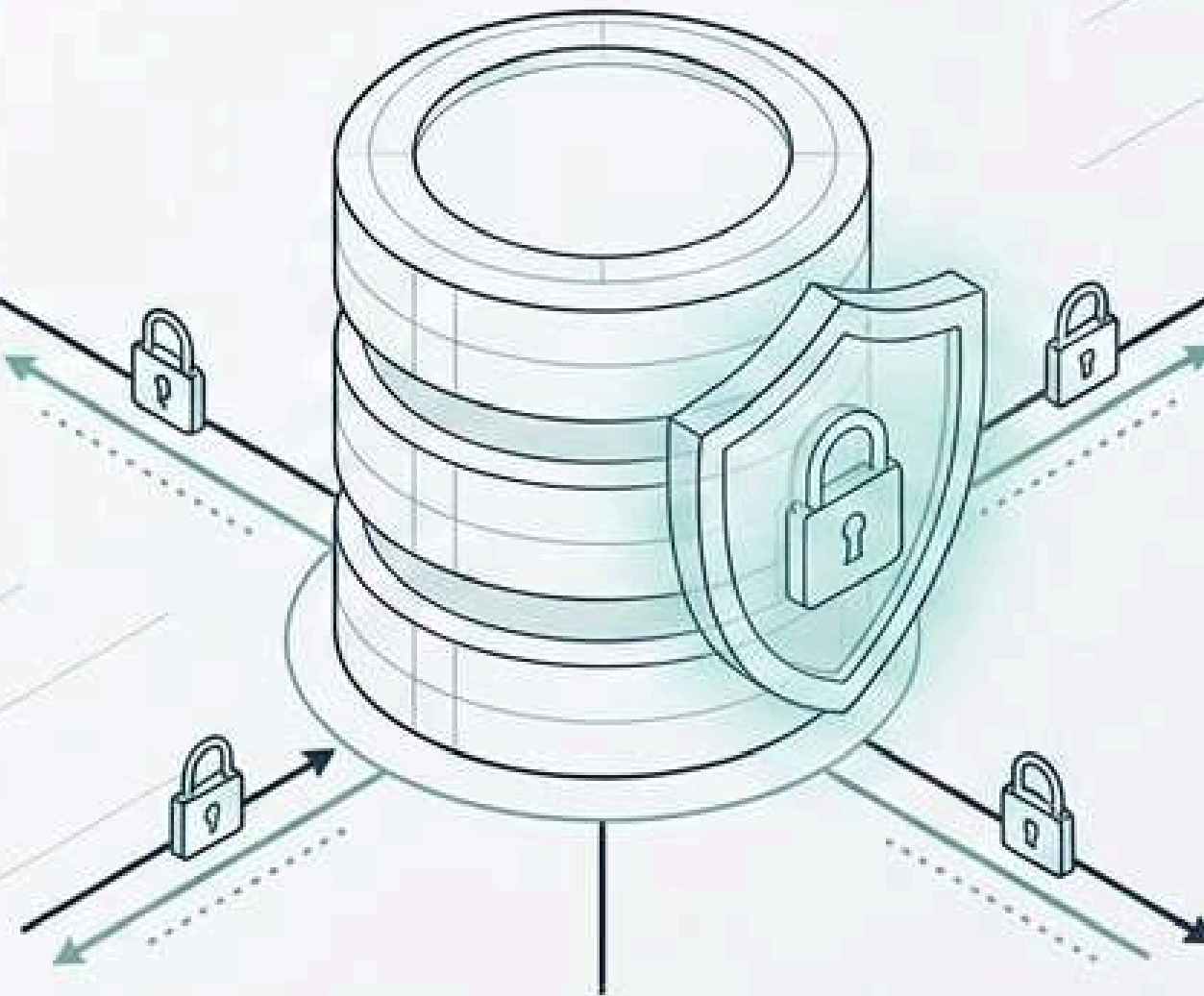
Secret Management

Environment secrets
encrypted in Vercel; strict
exclusion via .env.prod.



Automated Recovery

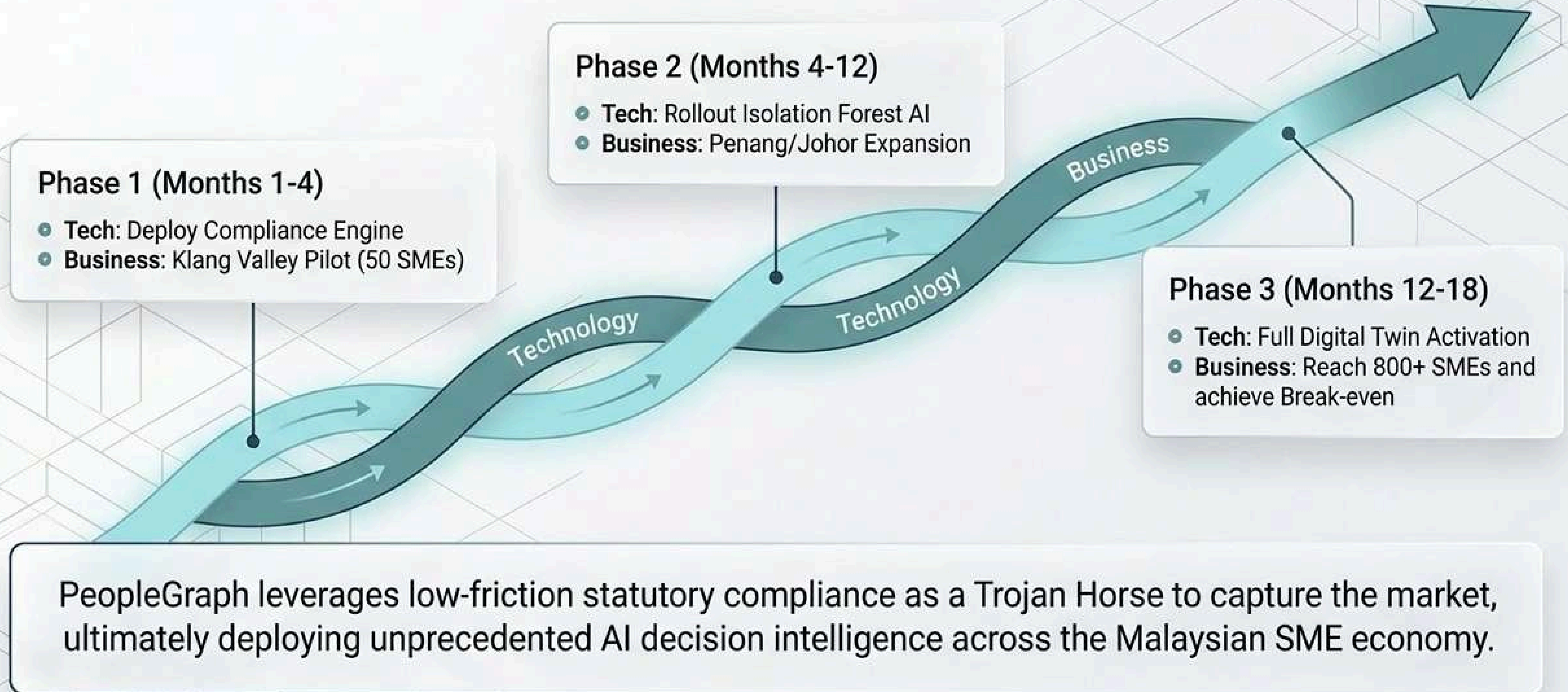
Multi-tiered PostgreSQL backup strategy
(Daily pg_dump, Pre-deployment
snapshots, 30-day rolling retention).



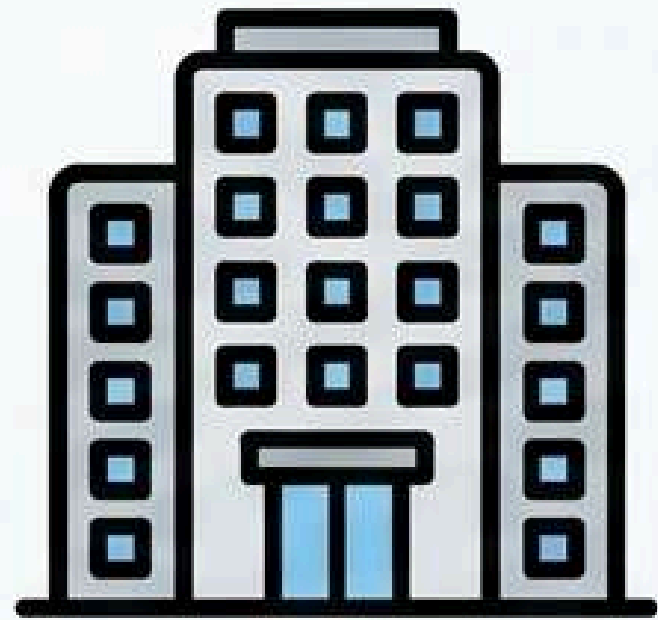
The SME HR Tech Landscape

	Compliance Automation	AI Decision Intelligence	Unstructured Data (Manglish) NLP	SME Pricing Fit
PeopleGraph	✔	✔	✔	✔
Payroll Platforms (BrioHR/Swingvy)	✔	✘	✘	✘
Enterprise ERPs (SAP/Workday)	✔	✔	✘	✘
Manual Excel	✘	✘	✘	✔

Synthesis: The Go-To-Market Execution



Our B2B Revenue Model: Enterprise Licensing



Enterprise Customers
(SMEs & Corporations)



Recurring License Fees
(Monthly/Annually)



PeopleGraph Platform Access
(AI Decision Engine & Digital Twin)

We generate **sustainable revenue** by offering the PeopleGraph platform as a subscription service, providing enterprises with continuous access to our AI-powered **decision engine** and **digital twin** technology for workforce optimization.